Boaz Meivar

PhD candidate, computer vision, Tel Aviv University · advised by Prof. Shai Avidan · Tel Aviv, Israel

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RESEARCH

I study when and why visual representations break: how object retrieval degrades with object scale and context, how to evaluate retrieval at web scale at a fraction of the cost, and whether vision-language models use the evidence in front of them.

Key results

- **State of the art on ILIAS-100M** instance retrieval (42.0 mAP@1000), training-free and without re-indexing the gallery.
- **Web-scale retrieval evaluation with 20× less indexing**, accurate to about 0.01 mAP.
- **Ill-posed tasks as a new lens on vision-language models:** when no single cue determines the answer, counterfactuals reveal which evidence a model actually uses.

EDUCATION

2024–now **PhD, Electrical Engineering** · Tel Aviv University
Research: AI and computer vision · advisor: Prof. Shai Avidan · expected 2028

2020–2022 **MSc, Aerospace Engineering** · Technion – Israel Institute of Technology
Advisor: Prof. Moti Karpel · thesis: data-driven aeroelastic flutter precursor

2008–2013 **BSc, Aerospace Engineering** · Technion – Israel Institute of Technology
Dean's List

EXPERIENCE

2022–now **Deep Learning Algorithm Researcher** · computer vision stealth startup
- Conceived and carried out the research, and built an MVP from scratch: retrieval of arbitrary objects across thousands of km^2 of geospatial imagery per search, from as little as one example.
- Combines self-supervised foundation models, active learning and gradient-boosted trees.
- The MVP became a cornerstone product of the company.
- Patent pending, first-named inventor.

2020–2022 **Computer Vision Algorithm Developer** · aerospace industry
Maintained and optimized a dim-target detection algorithm, wrote simulations in MATLAB, and ran operational analyses to improve its performance. Started and taught a 24-hour Python workshop for the engineering team.

2014–2020 **Dynamics and Loads Engineer** · aerospace industry
Multi-body dynamics simulations, calibrated with genetic optimization algorithms.

PUBLICATIONS

* equal contribution

Computer vision and machine learning

WACV 2027
accepted

Object-Only Is Not Context-Free: A Counterfactual Diagnostic for Context Removal in Object Retrieval
Boaz Meivar, Shai Avidan
KAP, the Keyhole Aperture Probe, measures how much an object's embedding depends on its surroundings, by revealing more of the scene around the object while the object region used for pooling stays fixed.

WACV 2026

Decoupling Shape and Texture in SAM-2 via Controlled Texture Replacement

Inbal Cohen*, Boaz Meivar*, Peihan Tu, Shai Avidan, Gal Oren

Fine-tuning with incrementally stronger texture replacement gives TextureSAM, which segments better where texture defines the boundary.

paper · arXiv:2505.16540 · code

arXiv 2026
under review

Beyond Resolution: Object-to-Image Ratio Mismatch in Instance Retrieval

Boaz Meivar, Ofir Kedem, Amit Edenzon, Gal Chechik, Shai Avidan

Object-to-image ratio mismatch, not resolution, explains over 80% of cross-distance retrieval loss for 9 of 12 backbones. Query-side scale augmentation reaches 29.2 mAP@1000 on ILIAS-100M, and adding an OWLv2 crop reranker gives a state-of-the-art 42.0, all training-free and without re-indexing the gallery.

arXiv:2610.11489

arXiv 2026
under review

Web Scale on a Budget: Tail-Aware Retrieval Evaluation

Boaz Meivar, Shai Avidan

From a 5% sample, a tail-aware (EVT) estimate lands within about 0.01 of full-scale mAP (MAE 0.005) across five backbones, validated at 100M images on ILIAS and INSTRE and on the full 759k-distractor GLDv2 pool.

arXiv:2610.17986

arXiv 2026
under review

Ill-Posed by Design: Probing Evidence Use in VLMs

Boaz Meivar, Shaked Perek, Shani Shvartzman, Eli Schwartz, Shai Avidan

Monocular metric object size is ill-posed, so counterfactuals separate unused evidence from redundant evidence. Across 12 open VLMs (3B to 397B) and six evidence channels, object identity is the strongest cue and global scene geometry is largely unused.

arXiv:2606.24335 · code

Machine learning for science and engineering

Nature Sci. Rep. 2026
Q1

Measurement-aware learning for reliable grain-boundary analysis in quantitative metallography

Boaz Meivar, Inbal Cohen, Matan Rusanovsky, Francis Quintal Lauzon, Julien Robitaille, Ofer Be'eri, Shai Avidan, Gal Oren

Removing global context sharply degrades detection of weak grain boundaries; MLOGRAPHY++ is validated on the grain-size measurement itself, not only on pixel accuracy.

doi:10.1038/s41598-026-58714-5

Annals BME 2026

Simultaneous 3D Quantification of Fluid–Structure Interaction in a Patient-Averaged Left Ventricle: A Lagrangian Experimental Framework

Omer Hadar*, Boaz Meivar*, Shai Avidan, Alex Liberzon

SAM-2 segmentation with 3D particle tracking gives co-registered, time-resolved 3D wall motion and intraventricular flow in a compliant heart model.

doi:10.1007/s10439-026-04293-5

AIAA J. 2025
Q1

Data-Driven Approach to the Development of an Aeroelastic Flutter Precursor

Boaz Meivar, Moti Karpel

A flutter early-warning system built end to end with essentially no prior knowledge of the aircraft; thresholds set on a wind-tunnel wing carried over to a flight test, warning up to 6 s before critical flutter onset.

doi:10.2514/1.J064327 · code

SKILLS

Methods	Self-supervised learning, few-shot retrieval, active learning, gradient-boosted trees, LoRA fine-tuning
Models	DINOv2, DINOv3, SigLIP2, SAM / SAM-2, OWLv2, open VLMs (Qwen-VL and InternVL families)
Engineering	Python, PyTorch, Hugging Face Transformers, FAISS; multi-GPU feature extraction and memory-mapped descriptor stores for 100M-image search
Also	Time-series analysis and anomaly detection, physics simulation and optimization; MATLAB
Languages	Hebrew, English